AIRCONNECT INDIA

Machine Learning-Based Analysis of Indian Airport Passenger Traffic and Connectivity

MDI4001 Machine Learning for Data Science

Version 2.0 | 19 October 2026

Team members: [Names and register numbers]

Problem statement

- Annual airport samples were too small for reliable supervised learning.
- A previous-year baseline outperformed the original Random Forest.
- Random train/test splits would leak future traffic patterns.
- The project needs classification, PCA, and a real confusion matrix.

Objectives

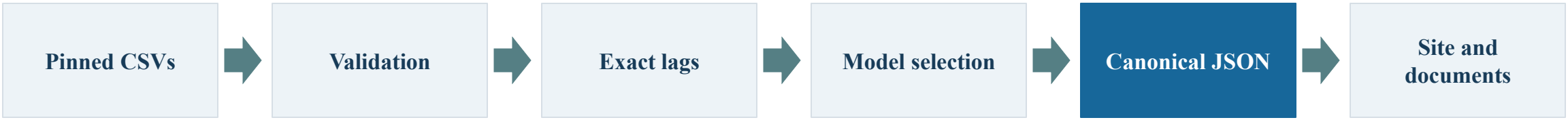
- Build an airport-month panel from validated DGCA-derived records.
- Predict next-month domestic passenger movements using only prior information.
- Classify at least 10% year-on-year growth with chronological evaluation.
- Evaluate K-Means groups and visualise the feature space with PCA.
- Publish consistent results in an offline website, report, and presentation.

Dataset and coverage

13,599	144	2015–2026	Pinned source
domestic airport-month rows	airports in monthly source	partial endpoints	commit + SHA-256

- Model window ends in December 2025. Partial 2026 is excluded.
- Targets remain observed. Missing calendar months never become zero.
- Complete domestic years reconcile between monthly and annual tables.

System architecture



Python prepares all results before the browser opens. The website has no runtime server, login, database, CDN, or cloud dependency.

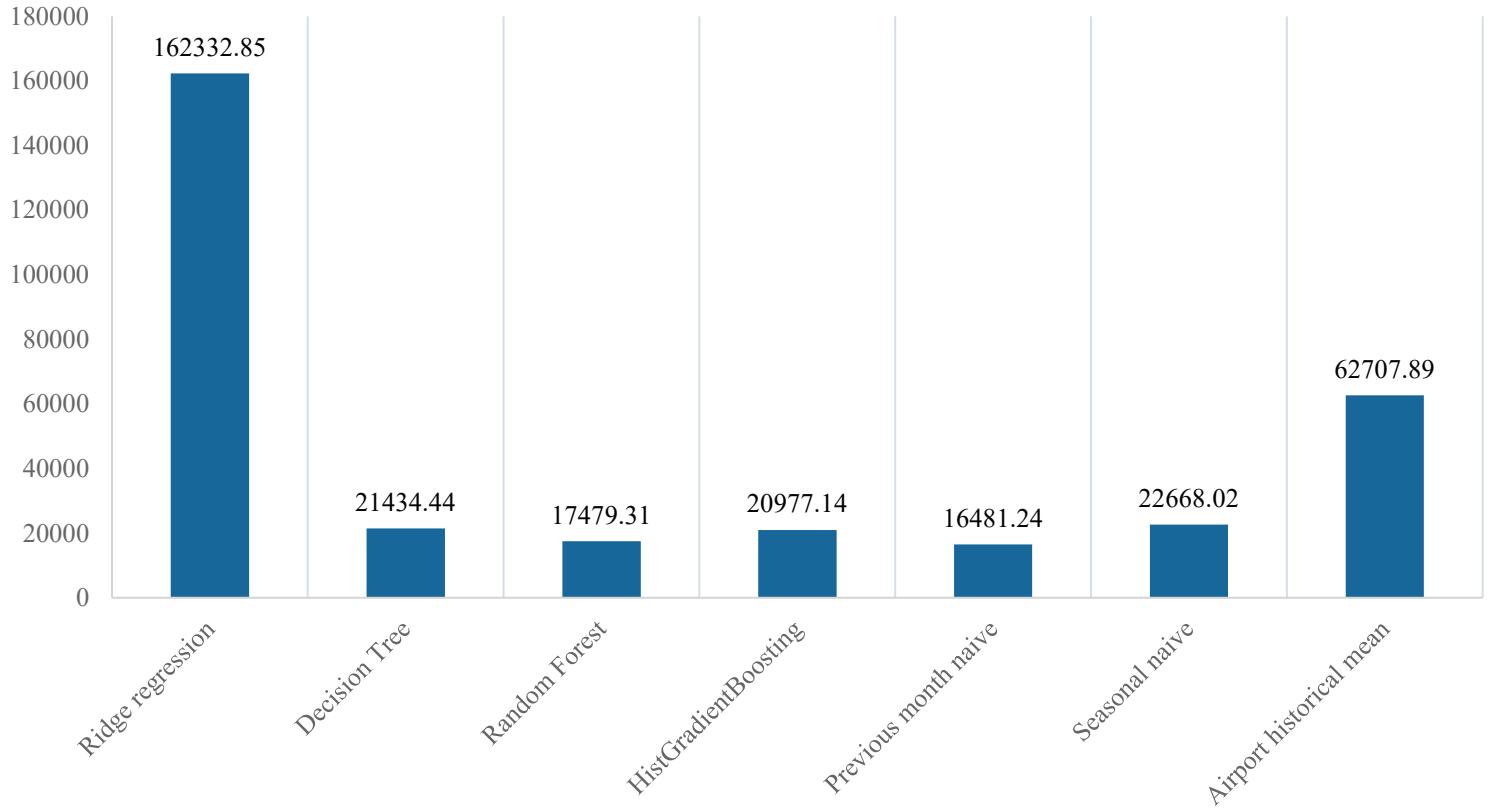
Data processing and feature engineering

01	**Exact lags: 1, 2, 3, and 12 months**
02	Shifted rolling means: 3, 6, and 12 months
03	Lagged growth, seasonal sine/cosine, airport identity
04	Train-only imputation and encoding inside each fold

Regression methodology

- Task: predict airport A in month T from information available before T.
- Candidates: Ridge, CART, Random Forest, Histogram Gradient Boosting.
- Baselines: previous month, same month previous year, airport historical mean.
- Expanding validation years: 2022, 2023, 2024. Final test: 2025.
- Selection metric: validation MAE. Reporting: MAE, RMSE, R^2, WAPE, median AE.

Regression results



Mean absolute error (passenger movements)

Selected ML

Random Forest

17,479 MAE

Best overall

Previous month

16,481 MAE

The baseline remains stronger on the untouched test.

Classification methodology

Class 1

Target traffic ≥ 1.10 × same month one year earlier

- Candidates: Logistic Regression, CART, Random Forest.
- Majority-class DummyClassifier provides the baseline.
- Validation balanced accuracy selects the classifier.
- Decision threshold 0.50. 2025 contains 588 positive cases from 1,357 rows.

Classification results

	Predicted 0	Predicted 1
Actual 0	688 True negative	81 False positive
Actual 1	196 False negative	392 True positive

Logistic Regression

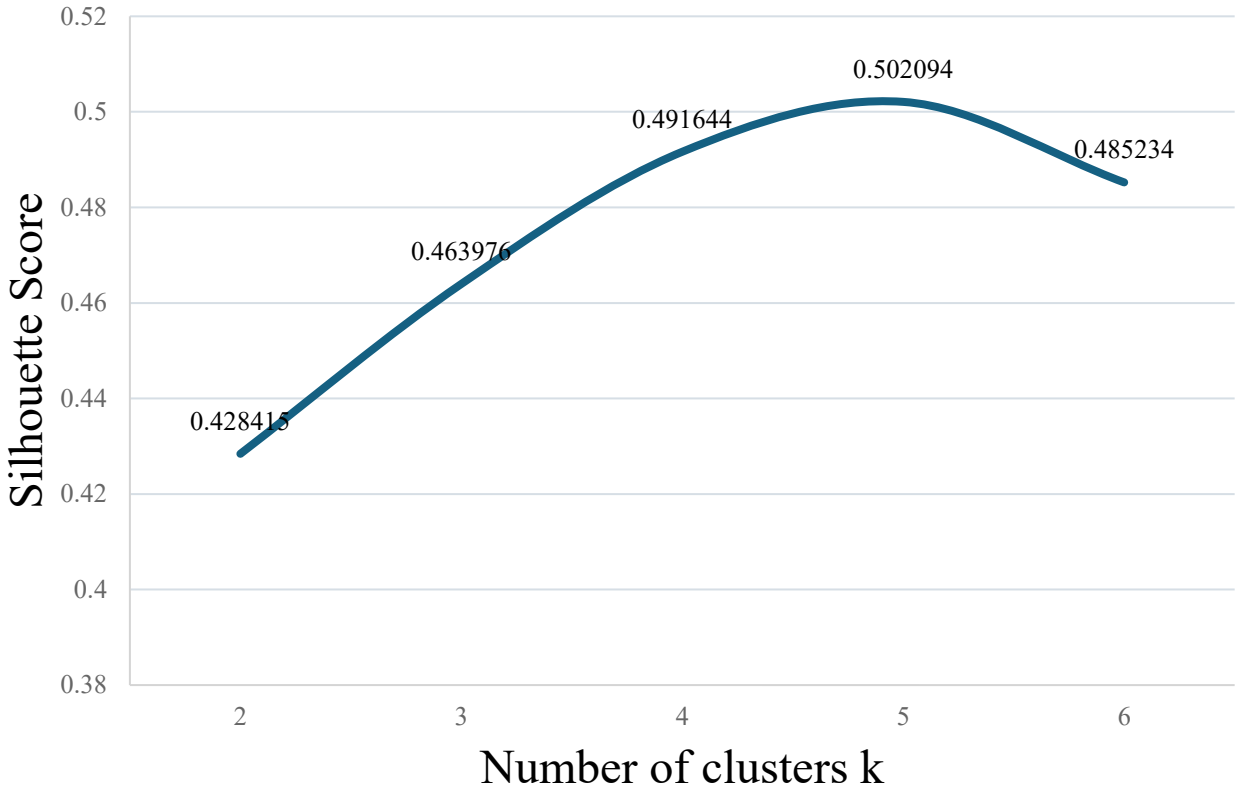
Balanced accuracy 0.781

F1 score 0.739

ROC-AUC 0.886

Random Forest scores higher on the final test, but validation selected Logistic Regression before the test was opened.

K-Means clustering



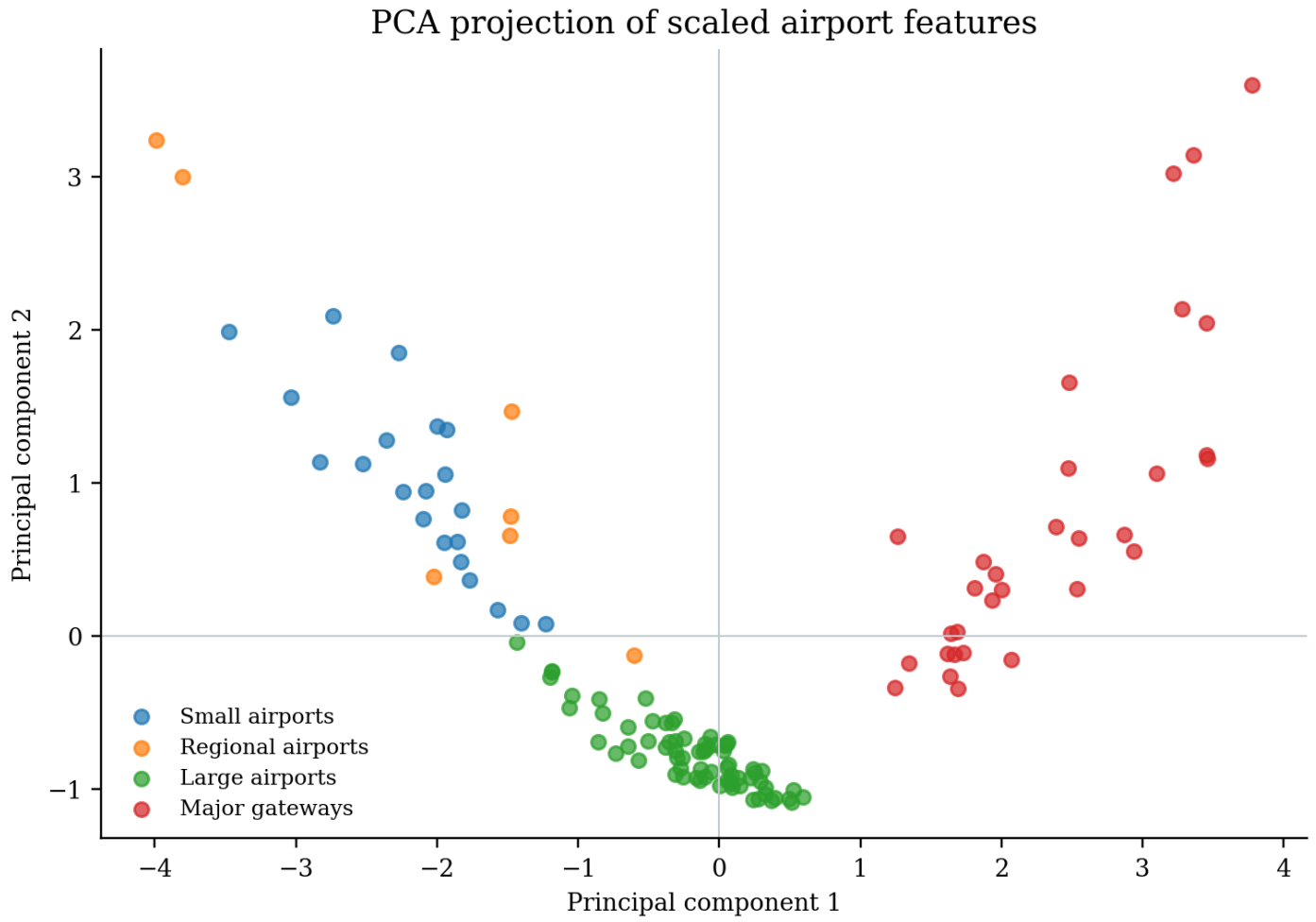
Selected k = 4

Silhouette 0.492

Small airports	21
Regional airports	7
Large airports	**70**
Major gateways	**31**

k = 5 has a slightly higher score but fails the minimum cluster-size rule.

PCA visualisation



PC1

44.85%

PC2

19.02%

Cumulative variance

63.87%

Offline website

Overview | **Airport Explorer** | **Machine Learning** | **Methodology**

- Regression, Classification, and Clustering/PCA sub-tabs
- Interactive SVG charts with keyboard-accessible data points
- Year, traffic group, airport search, map selection, and CSV export
- Viewport-aware tooltips for mouse, keyboard, and touch
- Static HTML, CSS, and JavaScript with local generated data

Findings and limitations

Findings

- Monthly data materially reduces trained-model error.
- The previous-month baseline remains the strongest forecast.
- Classification provides useful discrimination beyond the dummy.
- Four clusters balance separation and minimum group size.

Limitations

- One-step evaluation uses actual lagged observations.
- Pandemic disruption creates distribution shift.
- Continuity requirements affect eligible airport-months.
- Passenger counts do not measure capacity utilisation.

Conclusion and references

AirConnect India v2.0 demonstrates that careful temporal validation and strong baselines matter more than model complexity.

- Primary source: Kubicek (2026), India Aviation Stats, cleaned DGCA data.
- Methods: scikit-learn 1.7; deterministic seed 37.
- Full reproducibility: source hashes, canonical JSON, tests, website, report, and deck.

Repository github.com/kasinathca/airconnect